Less Than Equal Constraints

3 constraints

x1 ≤ z1, x2 ≤ z2, x3 ≤ z3

=>

```
[1 0 0] [x1]   [z1]
[0 1 0] [x2] ≤ [z2]
[0 0 1] [x3]   [z3]
```

Gx ≤ h

Greater-than converted

x1 ≥ z1, x2 ≥ z2, x3 ≥ z3

=>

```
[-1  0  0] [x1]   [-z1]
[ 0 -1  0] [x2] ≤ [-z2]
[ 0  0 -1] [x3]   [-z3]
```

Convert

x1 ≥ z1, x2 ≥ z2, x3 ≥ z3

TO

-x1 ≤ -z1, -x2 ≤ -z2, -x3 ≤ -z3

6 constraints, 3 variables

y1 ≤ x1 ≤ z1, y2 ≤ x2 ≤ z2, y3 ≤ x3 ≤ z3

=>

```
[-1 0 0]  [-y1]
[ 0 -1 0]  [-y2]
[ 0 0 -1]  [-y3]
[ 1 0 0]   [ z1]
[ 0 1 0] ≤ [ z2]  (6x3)
[ 0 0 1]   [ z3]
```

Gx ≤ h